Using Machine Learning to Train a Wearable Device for Measuring Students' Cognitive Load during Problem-Solving Activities Based on Electrodermal Activity, Body Temperature, and Heart Rate: Development of a Cognitive Load Tracker for Both Personal and Classroom Use

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Abstract

Automated tracking of physical fitness has sparked a health revolution by allowing individuals to track their own physical activity and health in real time. This concept is beginning to be applied to tracking of cognitive load. It is well known that activity in the brain can be measured through changes in the body's physiology, but current real-time measures tend to be unimodal and invasive. We therefore propose the concept of a wearable educational fitness (EduFit) tracker. We use machine learning with physiological data to understand how to develop a wearable device that tracks cognitive load accurately in real time. In an initial study, we found that body temperature, skin conductance, and heart rate were able to distinguish between (i) a problem solving activity (high cognitive load), (ii) a leisure activity (moderate cognitive load), and (iii) daydreaming (low cognitive load) with high accuracy in the test dataset. In a second study, we found that these physiological features can be used to predict accurately user-reported mental focus in the test dataset, even when relatively small numbers of training data were used. We explain how these findings inform the development and implementation of a wearable device for temporal tracking and logging a user's learning activities and cognitive load.